Exercise 1 - SQL Foundations

① Selecting & Sorting

Employees

Id	First_name	Last_name	department	Salary	Hire_date	City
1	Alice	Green	IT	70000	2020-01-10	Johannesburg
2	Brian	Lee	HR	45000		Cape Town
3	Cathy	Zulu	France	65000		Durban
4	David	Mokoena	Marketing	50000		Pretoria
5	Eva	Naidoo	IT	72000		Johannesburg

①-① Retrieve all columns

```
Select *
from employee;
```

①-② Find all unique ~~de~~department~~s~~

```
~~Select departmen~~
Select Distinct department
from ~~e~~employee;
```

Department
IT
HR
France
Marketing

① - ③ Retrieve first & Last name ordered by Salary descending

```
Select First_name,
       Last_name,
       Salary
From employee
order by salary DESC;
```

First_name	Last_name	Salary
Eva	Naidoo	72000
Alice	Green	70000
Cathy	Zulu	65000
David	Mokoena	50000
Brian	Lee	45000

① - ④ Retrieve the top 3 highest paid employees

```
Select Id,
       first_name,
       Last_name,
       Salary
From employee
order by salary DESC
Limit 3;
```

Id	First_name	Last_name	Salary
5	Eva	Naidoo	72000
1	Alice	Green	70000
3	Cathy	Zulu	65000

① - ⑤ Find employee in IT department

```
Select Id,
    First_name,
    last_name,
    department
From employee
Where department = 'IT';
```

Id	First_name	last_name	department
1	Alice	Green	IT
5	Eva	Naidoo	IT

① - ⑥ Find employee in finance with salary > 60000

```
Select Id,
  First_name,
  Last_name, salary,
  department
From employee
Where department = 'Finance' AND salary > 60 000;
```

Id	First_name	Last_name	Department	Salary
3	Cathy	Zulu	Finance	65 000

① - ⑦ Find employee in HR or Marketing

```
Select Id,
       First_name,
       last_name,
       department
From employee
Where department = 'HR' OR department = 'Marketing';
```

Id	first_name	last_name	Department
2	Brian	Lee	HR
4	David	Mokoena	Marketing

① - ⑧ Find employee not in IT

```
Select Id,
       First_name,
       Last_name
       department
From employee
Where department != 'IT';
```

Id	first_name	last_name	Department
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	David	Mokoena	Marketing

① - ⑨ Find employee in IT, HR or Finance using IN

```
Select Id,
    First_name,
    last_name,
    department
From employee.
Where department IN ('IT', 'HR', 'Finance');
```

Id	First_name	last_name	department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
5	Eva	Naidoo	IT

① - ⑩ Find employees in IT with salary > 65000 and city Johannesburg

```
Select Id,
    First_name,
    last_name,
    department;
    salary,
    city
From employee
Where department = 'IT' AND salary > 65000 AND
                                    city = Johannesburg
```

Id	First_name	last_name	department	salary	City
1	Alice	Green	IT	70000	Johannesburg
5	Eva	Naidoo	IT	72000	Johannesburg